

1 INVARIANT GENERATION

Segmented CoT review: large corpus → candidates
Historical-bug RAG: root-cause homologues
across middle-end and target backends
→ grounded, generalized invariants

2 CHECKER CONSTRUCTION

Invariant → static / dynamic checker
Applicable-ISA metadata
Four-state verdict + deterministic evidence
PASS · FAIL · NOT APPLICABLE · ERROR

confirmed root cause

3 AGENT LOOP

Orchestrator locks one invariant–checker pair
+ applicable ISAs + budget; Generator builds its seed
routing → build → coverage → checker
Feedback or C-Reduce + same-checker revalidation

EVIDENCE AND FEEDBACK

Every state and result → versioned blackboard
Revalidated violation → minimized PoC
Confirmed new finding, including a zero-day
→ next campaign's probe corpus